

By defn. of conditional, if $(p_1 \wedge p_2 \wedge \dots \wedge p_n)$ is True, then $(q \Rightarrow r)$ is True. By defn. of AND, if $(p_1, p_2, \dots, p_n)$ are all respectively True, then $(q \Rightarrow r)$ is True. $\therefore$ The argument form with premises $p_1, p_2, \dots, p_n$ and conclusion $q \Rightarrow r$ is valid. $\square$

12) Show that the argument form with premises $(p \wedge t) \Rightarrow (r \vee s)$, $q \Rightarrow (\neg t)$, $u \Rightarrow p$, and $\neg s$ and conclusion $q \Rightarrow r$ is valid by first using Exercise 11 and then using rules of inference from Table 1.

Soln: First we shall prove that the argument form with premises $(p \wedge t) \Rightarrow (r \vee s)$, $q \Rightarrow (\neg t)$, $u \Rightarrow p$, $\neg s$, and q and conclusion r is valid.

Step

1. $(p \wedge t) \Rightarrow (r \vee s)$

2. $q \Rightarrow (\neg t)$

3. $u \Rightarrow p$

4. $\neg s$

5. q

6. $\neg t$

7. u

8. t

9. p

10. $p \wedge t$

11. $r \vee s$

12. r

Reason

Premise

Premise

Premise

Premise

Premise

Modus ponens on 5, 2

Simplification on 6

Simplification on 6

Modus ponens on 7, 3

Conjunction on 9, 8

Modus ponens on 10, 1

Disjunctive syllogism on 11, 4

By Exercise 11, the argument form with premises $(p \wedge t) \Rightarrow (r \vee s)$, $q \Rightarrow (\neg t)$, $u \Rightarrow p$ and $\neg s$ and the conclusion $(q \Rightarrow r)$ is valid.

Second way

Premises: 1) $(p \wedge t) \Rightarrow (r \vee s)$ 2) $q \Rightarrow (\neg t)$ 3) $u \Rightarrow p$ 4) $\neg s$

Conclusion: $q \Rightarrow r$